

RFID

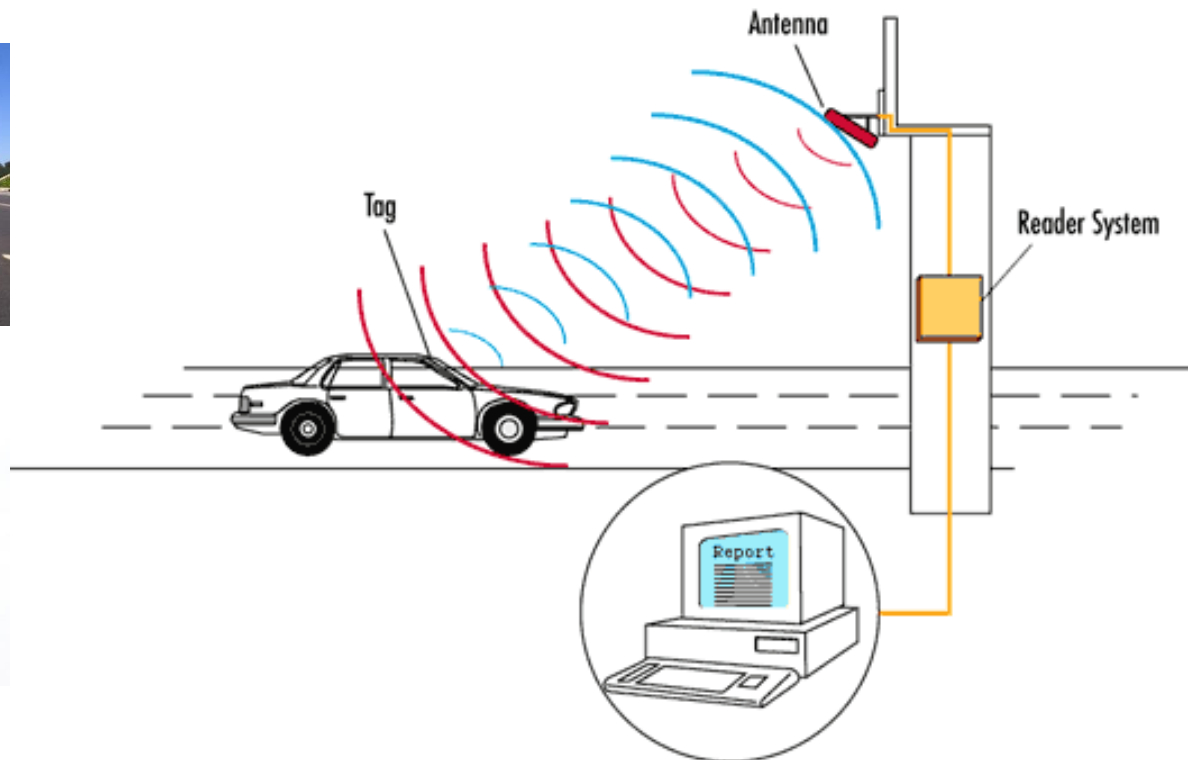
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RFID

- RFID means **Radio Frequency Identification**
- There are many types of RFID
- RFID tags can operate on different radio frequencies
- Main categories
 - **Active** – require a power source (e.g., a battery)
 - Example: Via Verde
 - Similar to a sensor node, but doesn't sense the environment
 - **Passive** – receive energy from a reader
 - Unlimited life time
 - Small size
 - Normally short-range
- A common RFID application is inventory management, to track stock levels in stores and at suppliers

Active Tags

- Use a internal power source (e.g., a battery)
 - Example: Via Verde

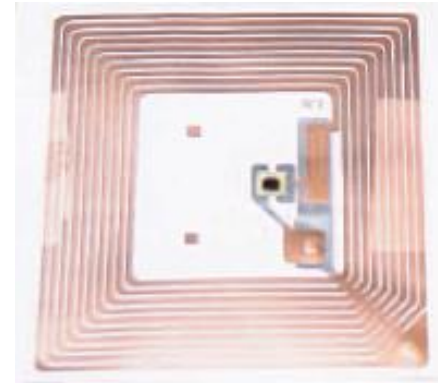


Passive Tags

- Main components
 - Antenna (power transfer and data communication)
 - Capacitor (energy storage)
 - Chip (data storage and transmission)
 - Encapsulation



128 kHz Trovan tag - encapsulated in a small glass vial



13.56 MHz Tiris tag - laminar plastic substrate with adhesive for easy attachment to goods

- Power transfer approaches
 - Magnetic induction
 - Electromagnetic (EM) wave capture

Data Transfer

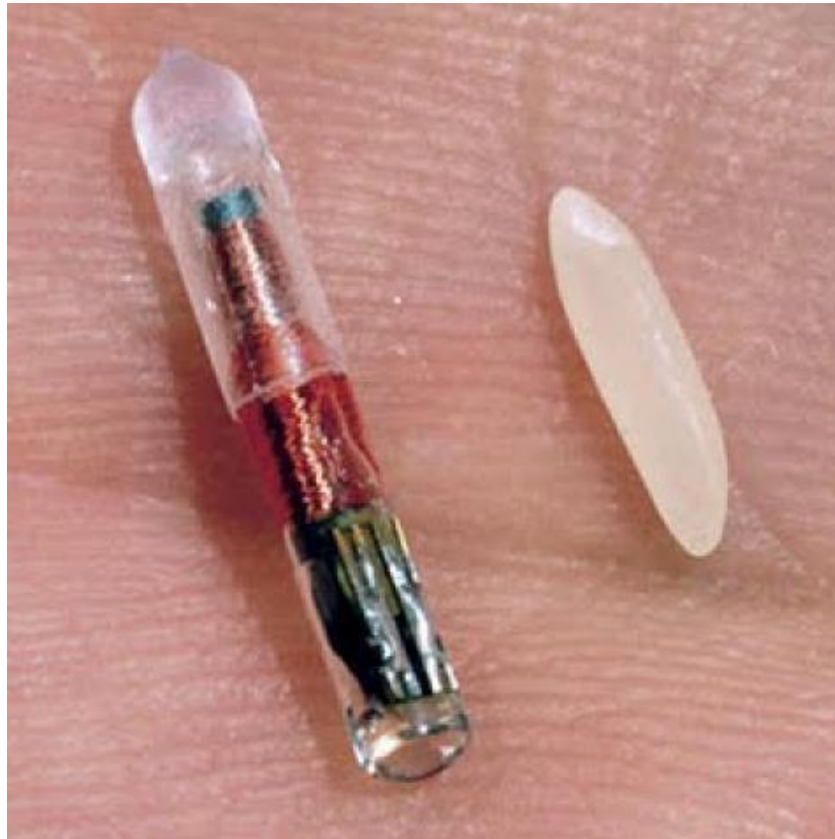
- Once powered by the reader, the chip
 - Synthesizes a digital signal to produce a modulation pattern
 - Send identity data back to the reader
- Multiple tags may be queried simultaneously by a reader
 - A simple form of MAC (randomization of the tag's reply time) may be used in order to minimize collision among tags

Types of RFID Tags

- In terms of data storage, there are two types of RFID
 - **Read-only tags**
 - Store a unique identifying number (ID), which is associated with more detailed product information in a database, probably on the Internet.
 - Any change of the product information have to be made in the database, as the data on the tag cannot be changed
 - Example: Pet ID chips
 - **Read/write tags**
 - These information stored in these tags can be overwritten
 - Therefore, it is not necessarily required Internet access to retrieve meaningful information from the tag
 - Example: Oyster card

Example: Read-only RFID Tag

- A read-only RFID tag of the type used in pet ID chips



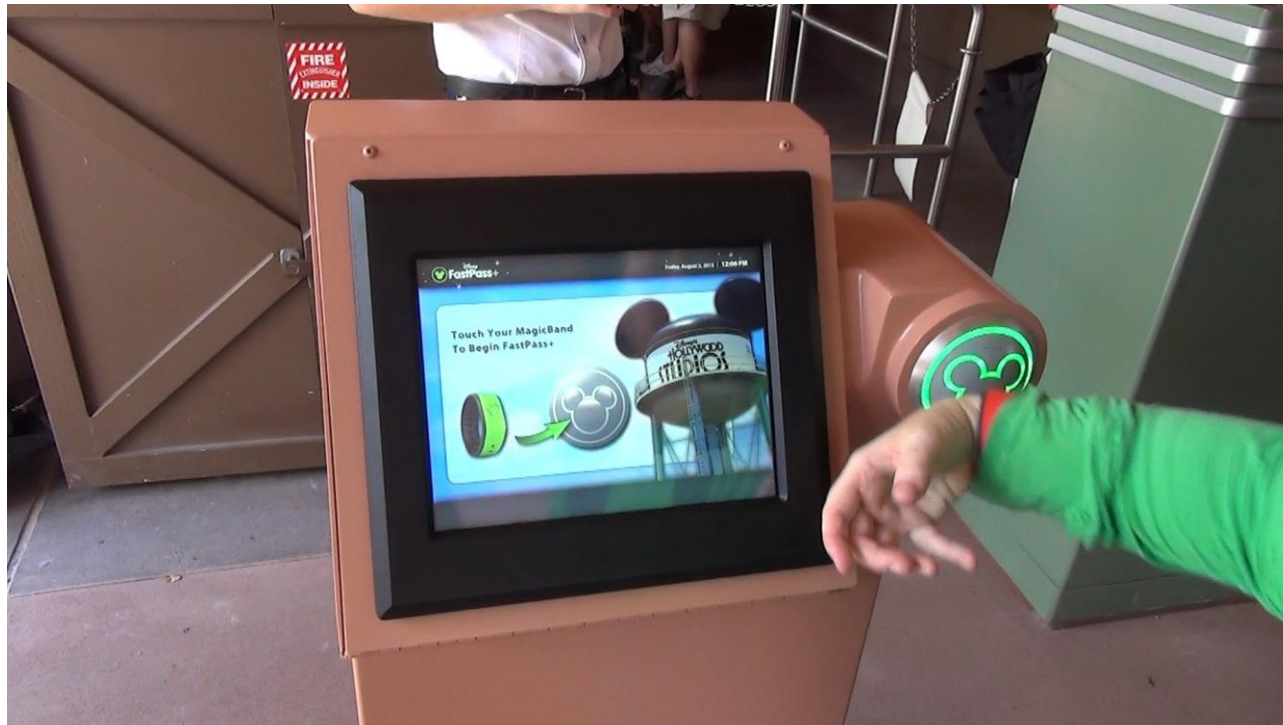
Example: Read/Write RFID Tag

- The London Oyster travelcard uses short-range read/write RFID to store local information about the cash or tickets stored on the card



Example: Read/Write RFID Tag

- Disney MagicBand wristband
 - Provides access to attractions and hotel rooms
 - Can be used to pay for purchases in the park
 - Tracks the movement of visitors around the parks, helping Disney better understand visitor behavior



Near field communication (NFC)

- A set of **short-range (4 cm)** RFID-based communication protocols and standards
- Uses the same standard as some RFID tags, but evolves it
- An **NFC-enabled device (e.g., a smartphone)** can behave both like
 - an **RFID tag** (and be read by a reader)
 - or like a **reader**
- Key advantage of NFC
 - RFID technologies uses many different data formats, so different RFID devices may not necessarily interoperate
 - **NFC provides a single common data format**—the NFC Data Exchange Format (NDEF), which allows all kinds of NFC-enabled devices and tags to share data

Smartphone NFC Applications

- Contactless payments
- Automation of tasks by configuring smartphone parameters **when touching a tag, using an app such as NFC Tools** (e.g., enable Bluetooth, disable Wi-Fi, enable GPS, and start Google Maps)
- Two-way **short-range communication between devices to share data** such as a photo or contact
- Streamline the process of **setting up a temporary high data-rate network connection** (e.g., a Bluetooth or Wi-Fi connection) **between two devices** (e.g., using Android Beam and Samsung S Beam)



RFID Sensing (Passive)

- Instead of an fixed ID, the **tag sends sensor data**
- Challenges
 - The sensor cannot use any power while the tag is not in communication with the reader (usual operating state)
 - Available energy is very small when the sensor is in reader range
 - Limits measurement techniques
- Application domains
 - Monitoring physical parameters
 - E.g., product spoiled by temperature, acceleration
 - Automatic product tamper detection
 - E.g, broken seal
 - Harmful agent detection
 - E.g., bacterial contamination of food products
 - Noninvasive medical monitoring
 - E.g., implanted sensors



QR Codes

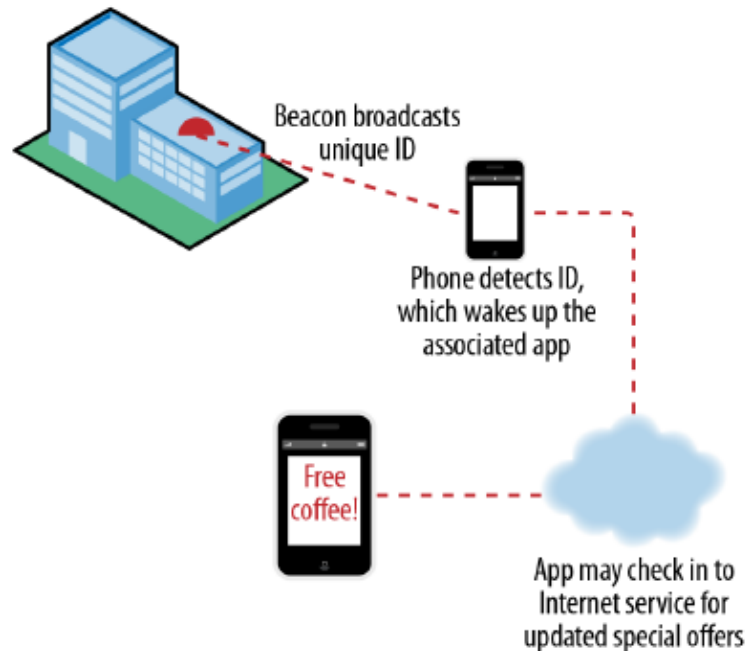
- Two-dimensional barcodes, which can be read by any imaging device able to extract the data encoded in the image, such as a smartphone
- Not a RFID technology, but similar trackable object
- Like read-only RFID tags, QR codes rely on the code being mapped to more extensive product information/functionality in a database

Beacons

- Another type of trackable object
- Longer range than passive tags and barcodes
- Example: Apple iBeacons
- Used to provide services based on the location information
- Use Bluetooth Low Energy devices to broadcast a unique ID that can be received by nearby Bluetooth-enabled devices, such as smartphones
- The smartphone looks up the beacon's ID in an online database, which provides additional information such as who owns the beacon, and exactly where it is
- The strength of the radio signal between the beacon and phone may help determine how far the phone is from the beacon

Beacons

- Beacon-enabled apps on the smartphone can then use that ID to provide contextually relevant functions/information, for example:
 - Entering a favorite department store might trigger the loyalty app on your phone to notify you of a special offer or discount voucher
 - Passing a beacon in the cleaning products aisle of the grocery store might trigger a shopping list app to remind you to pick up dishwasher detergent
 - A beacon in your car might trigger your phone to bring up the option to unlock the car



Bibliography

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